Problem Formulation

Define two classes: *class I* and *Class II*
let d(X) be a discriminant function = $W^T X$
if X ∈ *class I* then d(X) > 0
if X ∈ *class II* then d(X) < 0

For misclassified *X* :
If X ∈ *class I* and misclassified as Class II then d(X) will be <0
If X ∈ *class II* and misclassified as Class I then d(X) will be >0

Define Error function as:
Error = $\sum_{\text{sclassified vectors (i)}} W^T X_i \Rightarrow (W^T X_i)$ take positive and negative values *X* **x**

Define δ_i as :
If X ∈ *class I* and misclassified as Class II then d(X) will be <0 ⇒ δ_i = -1 ⇒ d(X) δ_i >0
If X ∈ *class II* and misclassified as Class I then d(X) will be >0 ⇒ δ_i = +1 ⇒ d(X) δ_i >0